📖 MULTI-OTO: Comprehensive User Manual

1. Introduction

Welcome to **MULTI-OTO**. On the surface, this plugin is a pristine, Phase-Aligned Multiband Upward/Downward Compressor with ADAA (Anti-Derivative Antialiasing) saturation. Beneath the surface, it is a **Fractal Dynamics Engine** capable of cascading up to 128 multiband compression nodes in series.

Whether you need to gently glue a vocal bus, perfectly balance a master track, or utterly obliterate a sine wave into an infinite, glitching, cybernetic texture—MULTI-OTO is designed to handle both extremes with mathematical precision and zero CPU overload.

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2. Global Architecture

The signal flow of MULTI-OTO operates as follows:

Input → Pre-Drive (ADAA Saturation) → [Crossover → Upward/Downward Comp → Resynthesis] x (2 to 128 Cascades) → Post-Filters → Master Limiter → Output.

The plugin is divided into two main processing halves: **Stage 1** (handles the first half of your selected cascades) and **Stage 2** (handles the second half). This allows you to aggressively flatten transients in Stage 1 and extract microscopic textures in Stage 2.

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3. Module Breakdown

PRE-DRIVE & X-OVER

This section conditions the signal before it enters the OTT cascade.

- **ON (Bypass):** Bypasses the Pre-Drive saturation block.
- **IN (Input Gain):** Adjusts the global input level (-24dB to +24dB).
- **DRIVE:** Controls the amount of signal fed into the ADAA saturator.
- **ODD:** Blends in Odd-order harmonics (generates aggressive, square-wave-like bite and presence).
- **EVEN:** Blends in Even-order harmonics (adds warm, asymmetrical, tube-like thickness).
- **COUNT:** The heart of the plugin. Selects the total number of OTT nodes the signal will pass through (**2, 4, 8, 16, 32, 64, or 128**).
- **LOW X:** Sets the crossover frequency separating the Low and Mid bands (20Hz – 1000Hz).
- **HIGH X:** Sets the crossover frequency separating the Mid and High bands (1000Hz – 20000Hz).

STAGE 1 & STAGE 2

Each Stage governs exactly half of your total **COUNT** (e.g., if COUNT is 64, Stage 1 controls nodes 1-32, Stage 2 controls nodes 33-64).

- **ON (Bypass):** Bypasses the respective Stage entirely.
- **LOW G / MID G / HI G (Gain):** Post-compression makeup gain for each specific frequency band (-24dB to +24dB).
- **LOW D / MID D / HI D (Depth):** Scales the intensity of the Upward and Downward compression for each band (0% to 100%).
- **TIME (Macro Time):** A global multiplier for all Attack and Release times in this stage. At 100%, A/R times are exact. Lower values speed everything up; higher values slow everything down.
- **MIX:** Parallel processing per stage. Blends the compressed signal with the uncompressed signal entering that specific stage (0% to 100%).
- **ATK L / ATK M / ATK H (Attack):** The time it takes for the compressor to react to transients for each band.
- **REL L / REL M / REL H (Release):** The time it takes for the compressor to recover. *Note: Staggering these times across bands creates the iconic "sweeping" spectral tail.*

🎛️ MASTER

Final polish, phase behavior, and absolute safety clipping.

- **HPF / LPF:** High-Pass and Low-Pass filters applied *after* the entire 128-cascade engine to clean up extreme sub-rumble or harsh high-end noise.
- **PHASE MODE:**
  - **ALIGN PHASE:** Compensates for all crossover phase shifts using All-Pass filters. Ensures the Dry/Wet mix is perfectly flat with zero comb-filtering.
  - **COLOR PHASE:** Bypasses phase compensation. Passing through up to 128 uncompensated crossovers creates massive phase dispersion (group delay). *Crucial for extreme sound design.*
- **MIX (Dry/Wet):** Global blend between the original input and the fully processed signal.
- **CEIL (Limiter Ceiling):** A hard-clipping safety net (-2.0dB to -0.1dB). Because 128 upward compressors can generate immense gain, this zero-latency limiter ensures your speakers and ears remain safe.
- **OUT (Output Gain):** Final volume adjustment (-24dB to +24dB).

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4. Use Cases: The Dual Nature of MULTI-OTO

🟢 Scenario A: The "Normal" (Conventional / Surgical) Use

Use MULTI-OTO as a mastering-grade multiband dynamics tool to add pristine presence to vocals, drums, or master buses without destroying the mix.

- **COUNT:** `2` or `4` (Keeps the CPU load minimal and the effect transparent).
- **PRE-DRIVE:** `OFF` or minimal (e.g., Drive 10%, Even 20%) for gentle warming.
- **PHASE MODE:** `ALIGN PHASE` (Absolutely critical. Ensures that blending the Dry/Wet knob won't cause hollow comb-filtering).
- **STAGE 1:** Depth 30-50%. Slow down the `TIME` macro to 150% to retain natural transients.
- **STAGE 2:** `OFF` (You only need one stage for standard processing).
- **MIX:** 20% to 40% (Classic parallel compression technique).

Result: A perfectly phase-coherent, beautifully glued signal that lifts the hidden details of a room reverb or breath without coloring the original transient.

🔴 Scenario B: The "Eccentric" (Freakish / Chaos) Use

This is what MULTI-OTO was truly built for. Use this on simple waveforms (Sine/Saw) or foley samples to generate Neurofunk neuro-basses, Color Bass squelches, and infinite cybernetic glitches.

- **COUNT:** `64` or `128` (Unleash the full fractal cascade).
- **PHASE MODE:** `COLOR PHASE` (Passing through 128 Linkwitz-Riley crossovers causes the phase to twist up to 46,000 degrees. This stretches transients into "pew-pew" laser sounds and watery squelches).

- **PRE-DRIVE:** Drive 80%, Odd 100%. (Crush the initial signal into a square wave to feed the OTT nodes maximum harmonics).
- **STAGE 1 (The Flattener):** DEPTH 100% on all bands. TIME 20% (Lightning fast). This completely pulverizes the input into a flat block of sound.
- **STAGE 2 (The Texture Extractor):**
  - **LOW D:** 0% (Crucial! Keep the sub-bass clean so the mix doesn't turn into pure mud).
  - **MID D / HI D:** 100%
  - **Releases:** Make REL H very fast (40ms) and REL L very slow (250ms).
- **The Magic (Infinite Tail):** MULTI-OTO continuously injects an inaudible -144dB stereo dither into the engine. When the input audio stops, the 128 Upward Compressors instantly grab this microscopic dither and amplify it to the maximum +36dB range limit. Because the Highs release faster than the Lows, the resulting noise floor morphs into a massive, stereo-widened, infinitely sweeping "Schwaaaaaa" tail.

Result: A simple drum loop or sine wave is instantly transformed into a complex, stereo-glitching, hyper-compressed wall of sound with an evolving tail that never truly dies.

Enjoy the absolute control, and embrace the infinite chaos.